3.	$(C_V - R \quad R/2)$	1.0
합계		**3.0**

문제 1C (3.0점)

,

$$\varphi \sim k\frac{q}{R} \sim \alpha\frac{\rho R^3}{R} \sim \alpha\rho R^2, \tag{1}$$

α .

. R

$+\rho$, r, -2ρ .

$$\alpha\rho R^2 + \alpha(-2\rho)r^2 = 0, \tag{2}$$

$$\frac{R}{r} = \sqrt{2}. \tag{3}$$

채점 기준

№	평가 항목	배점
1.	$\varphi \sim \rho R^2$	1.0 1.0 - 0.5 0
2.		1.0
3.		1.0
합계		**3.0**

문제 2. 퍼텐셜 에너지로 본 평형 (10.0점)

1. 도입 (1.0점)

1.1 **[1.0점]** 액 .

$$\Delta U_S = -(\sigma_2 - \sigma_1)\Delta S. \tag{1}$$

Δl

$$(\sigma_2 - \sigma_1)\Delta l = \sigma_0 \Delta l \cos\theta. \tag{2}$$

(1) (2)

$$\Delta U_s = -\sigma_0 \cos\theta \ \Delta S. \tag{3}$$

2. 수직 원통형 관 안의 물 (2.0점)

2.1 **[0.5점]** .

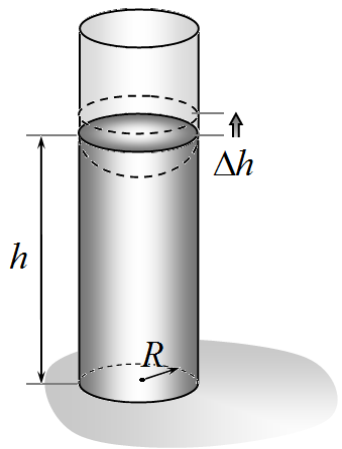
ΔU_S

Δh

$$\Delta U_S = -\sigma_0 \cos\theta \cdot \Delta S = -\sigma_0 \cos\theta \cdot 2\pi R\Delta h, \tag{4}$$

$\Delta S = 2\pi R\Delta h$

.



2.2 **[0.5점]** ΔU_G

Δh

$$\Delta U_G = \pi R^2 \Delta h \rho g h. \tag{5}$$

$$\Delta m = \pi R^2 \Delta h \rho$$

, h .